

ME 165
Basic Mechanical Engineering

Lecture 06

Structure

Sadia Tasnim

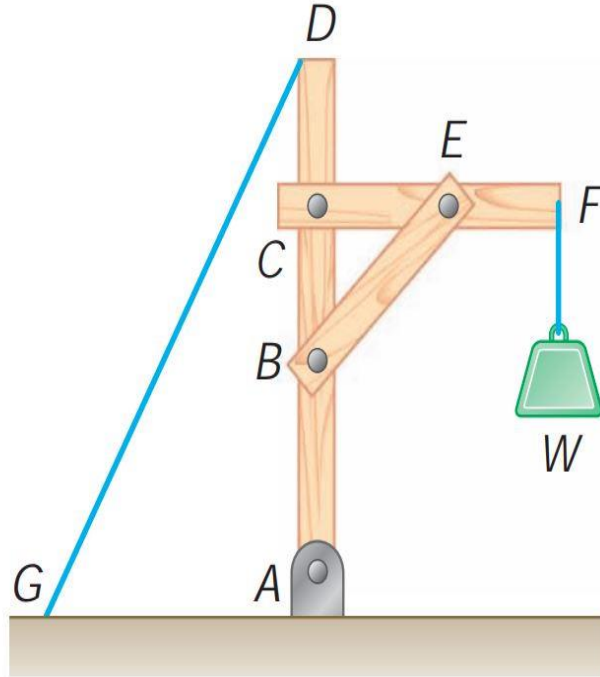
Lecturer

Department of Mechanical Engineering, BUET

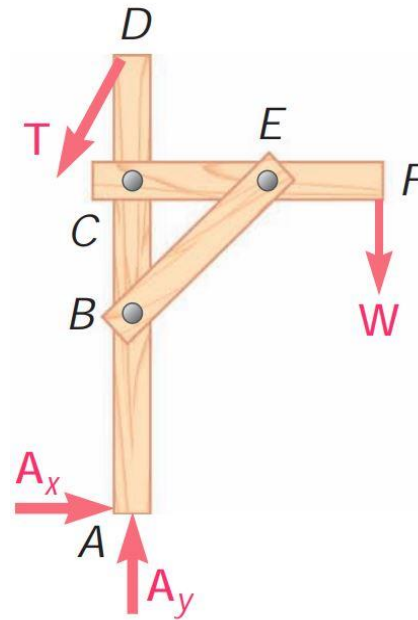
Analysis of Structures

Structure:

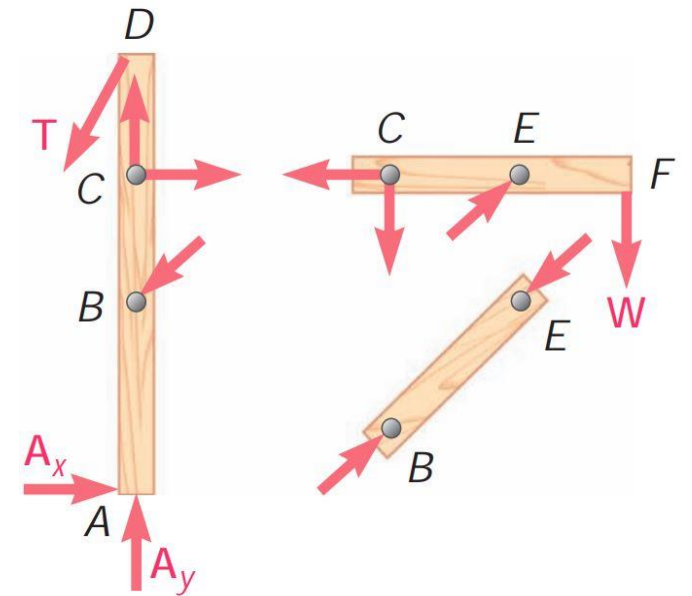
- Consists of several connected parts or members.
- Apart from the external forces acting on the structure, we have to consider internal forces that hold together the various parts of the structure.



A Structure



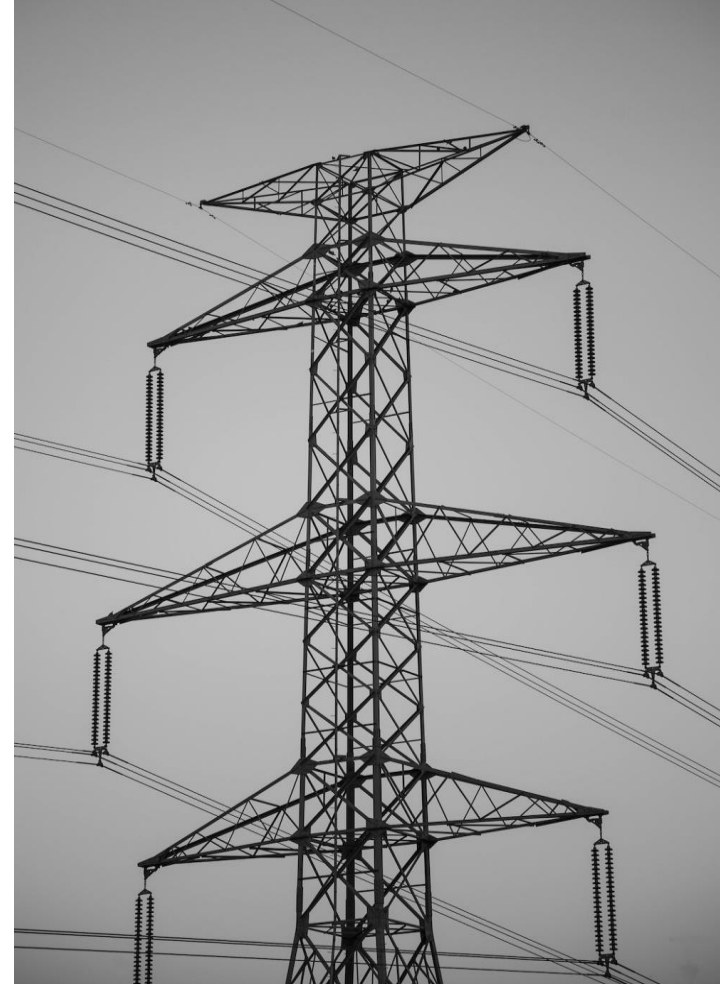
FBD of the structure



FBD of individual members

Types of Structures

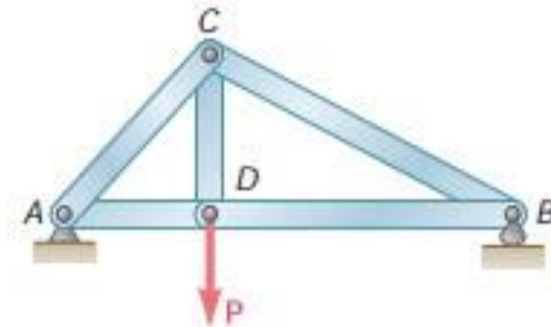
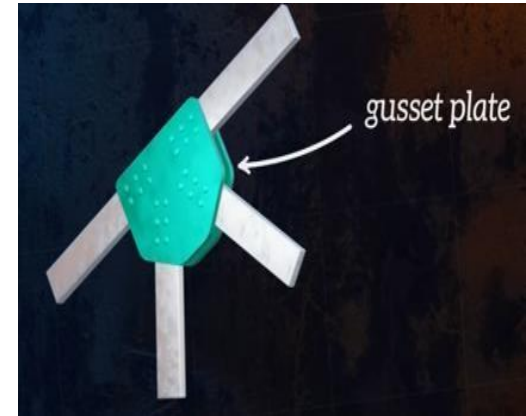
Truss:



Types of Structures

Truss:

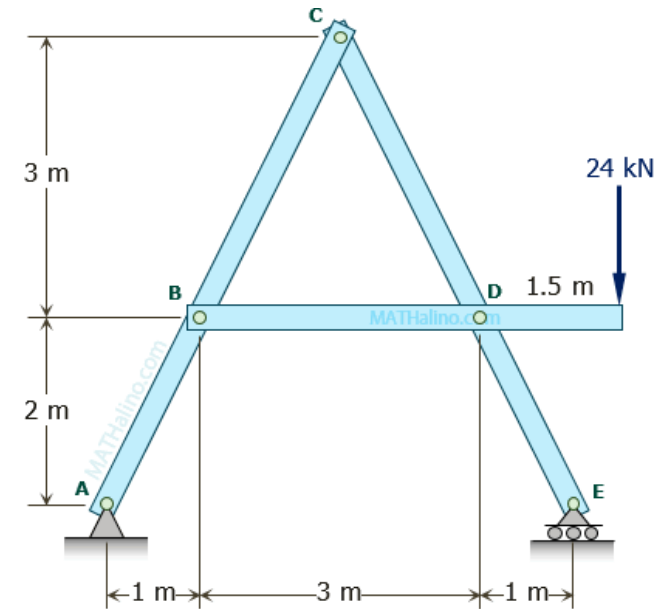
- A rigid structure which is designed to support loads
- Consists of exclusively of straight members
- Connected at joints located at the ends of each member by **pinned connection**
- Loads are only ever applied at the joints of the truss; but **not in the middle of the member**
- All the members are **two-force members**
- Stationary
- Fully constrained



Types of Structures

- **Frame:**

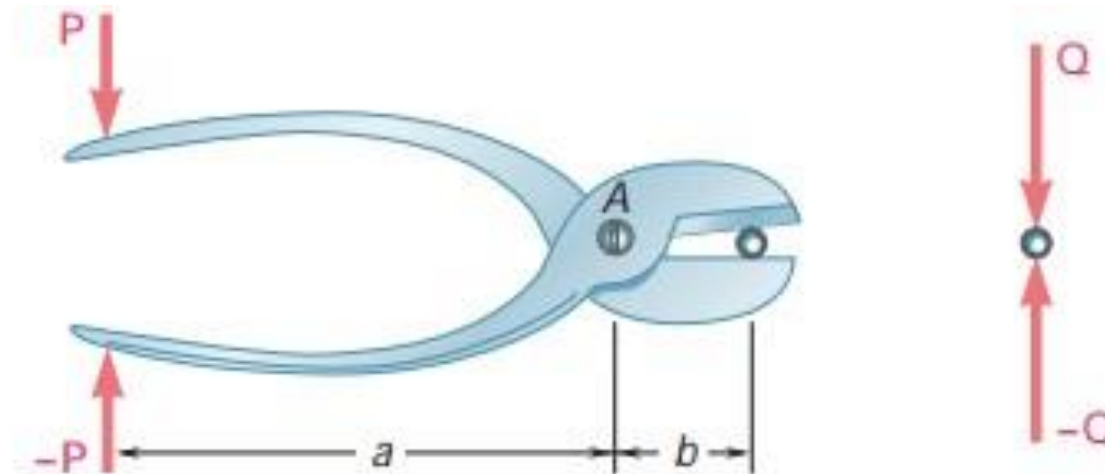
- Almost all the characteristics of a truss are valid for frame. But there are extra features in frame which differentiate it from truss.
 - Contains **at least one multi-force member** i.e., a member acted upon by three or more forces which, in general, are not directed along the member
 - Can also have **loads applied directly to its members**
 - **Not necessarily always have pin joints**



Types of Structures

Machine:

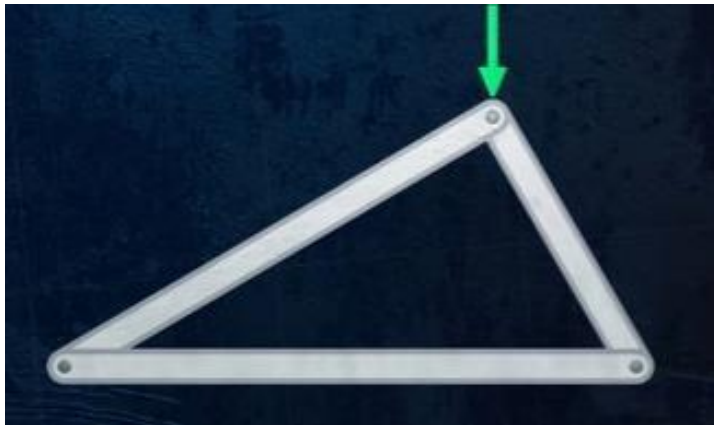
- Are designed to transmit and modify loads
- Have moving parts
- At least one member is multi-forced member



Types of Structures

Truss:

The base shape of truss is three members connected to form a triangle, a stable shape which won't deform when loads applied to it.



(a) Stable



(b) Unstable

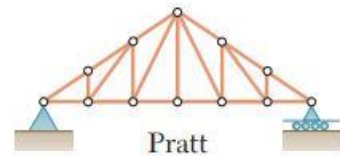


(C) Stable

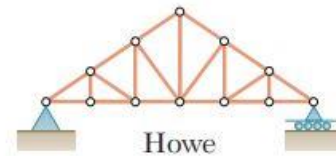
Joining four members together does not form a stable structure. Because of the application of load, angles between members can change without any change in the length of the members.

Types of Structures

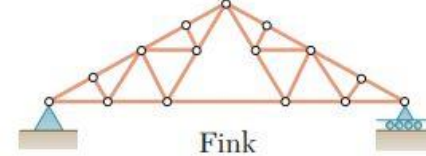
Truss:



Pratt

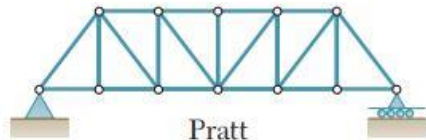


Howe

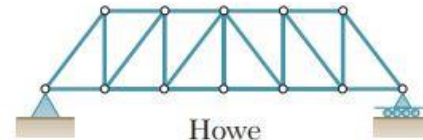


Fink

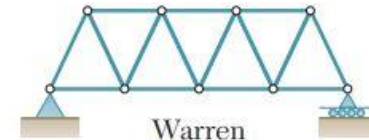
Typical Roof Trusses



Pratt



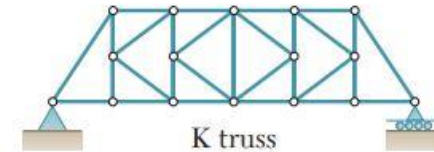
Howe



Warren

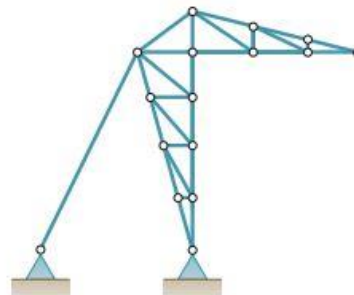


Baltimore

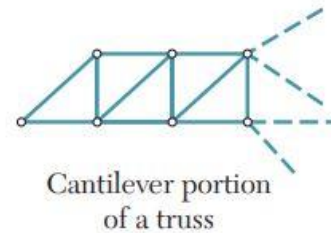


K truss

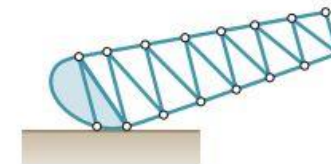
Typical Bridge Trusses



Stadium



Cantilever portion
of a truss

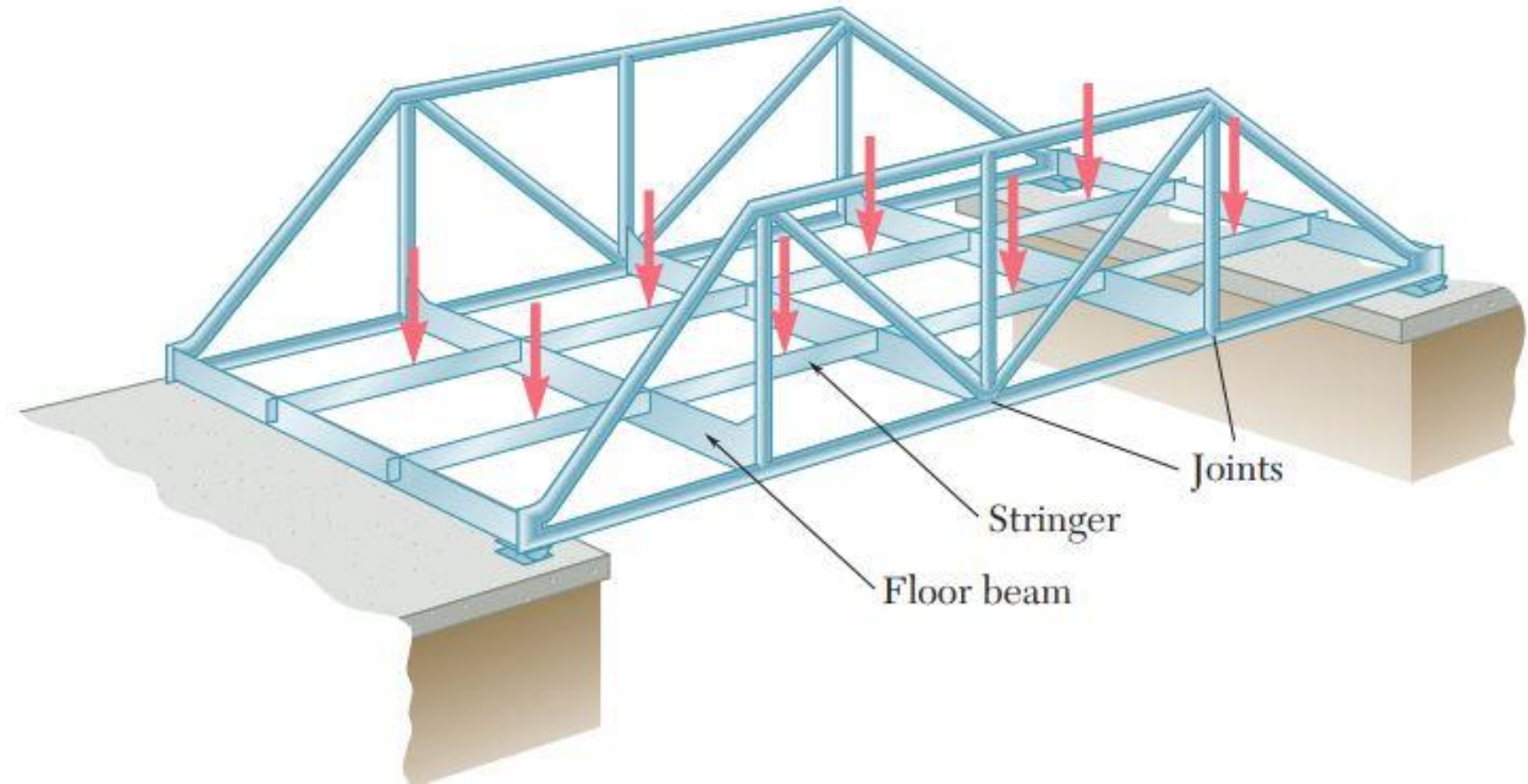


Bascule

Other Types of Trusses

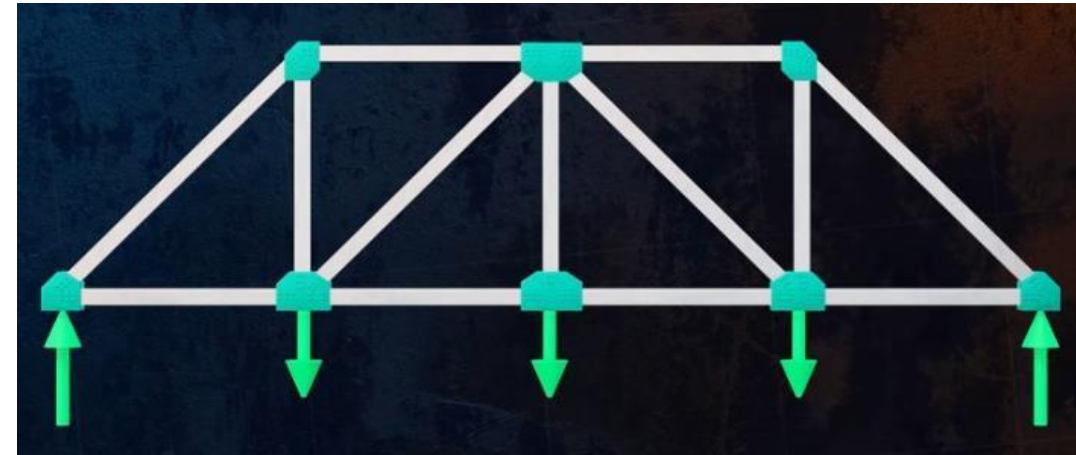
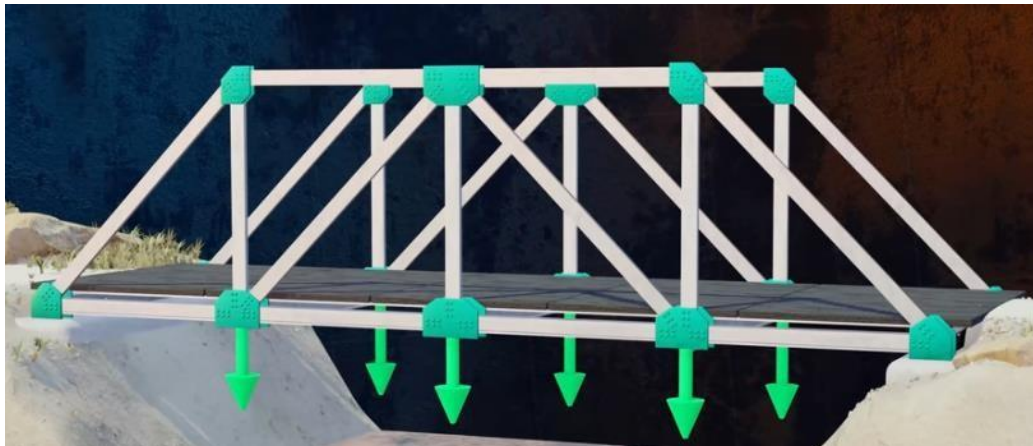
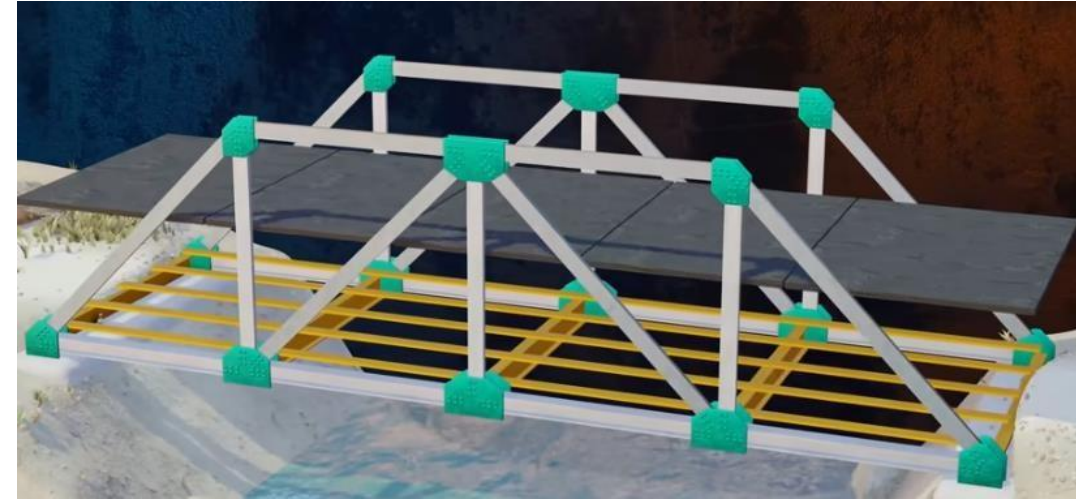
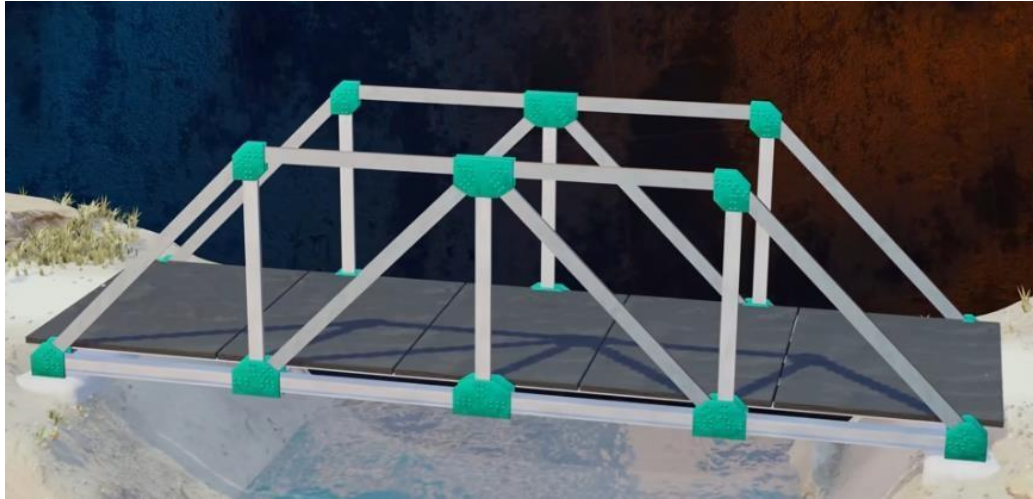
Types of Structures

Truss:



Types of Structures

Truss:



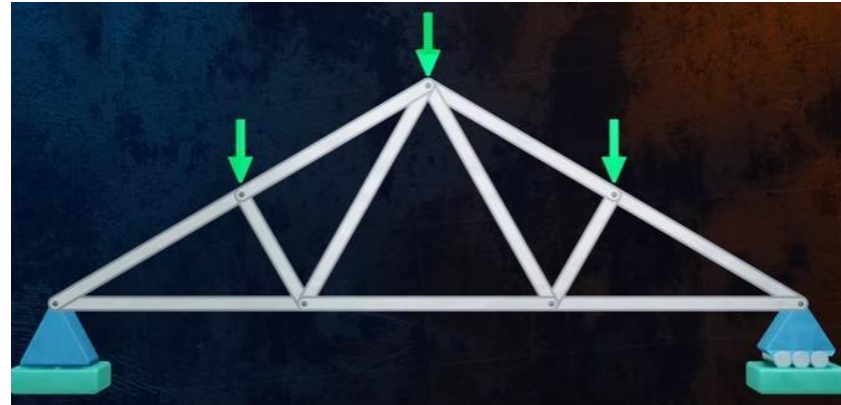
Analysis of Trusses by Method of Joints

- **Zero force members:** Members don't carry any loads at all
 - 2 main configurations where we have zero force members:
 - i) When three members connected at a single joint such that
 - two of the members are aligned / collinear
 - third member makes an angle with other two
 - no external loads or support reactions applied to that joint

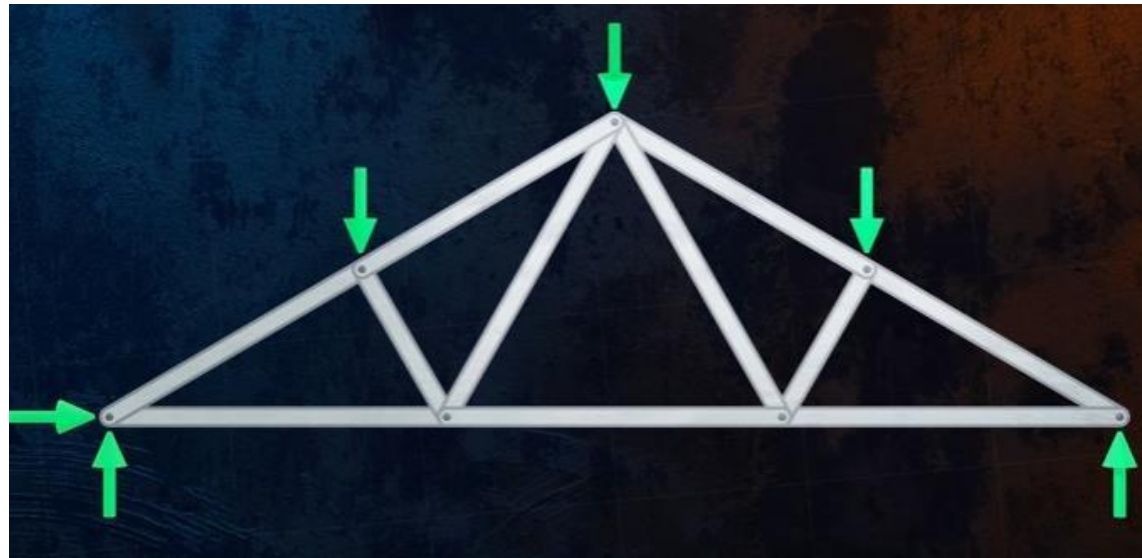


- ii) When two members connected at a joint such that
 - two members are not aligned / non collinear
 - no external loads or support reactions applied to that joint

Analysis of Trusses by Method of Joints

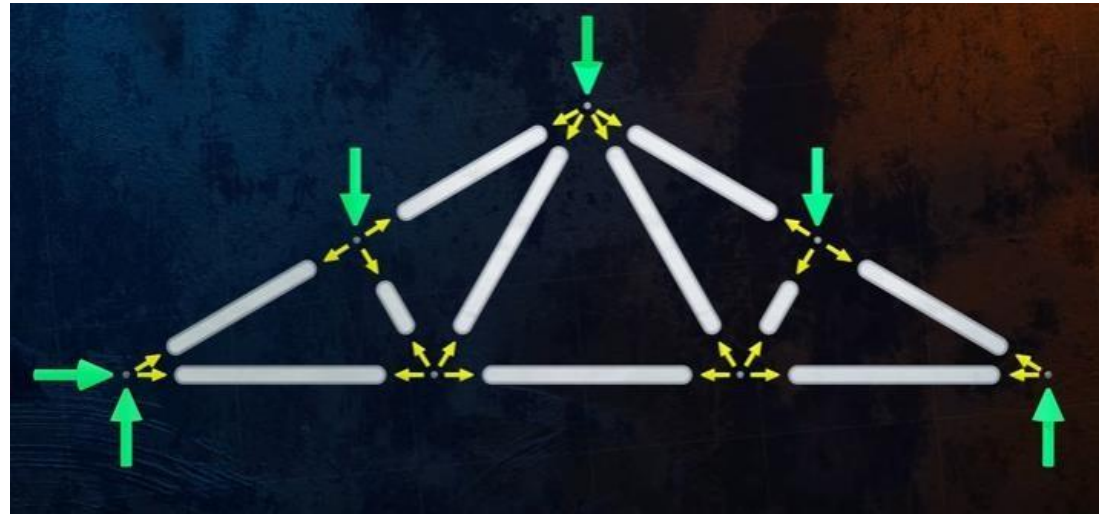


- 1) Draw free body diagram of **the truss** and solve **reaction forces**.



Analysis of Trusses by Method of Joints

2) Draw free body diagram of **each joint** and solve **internal forces**.



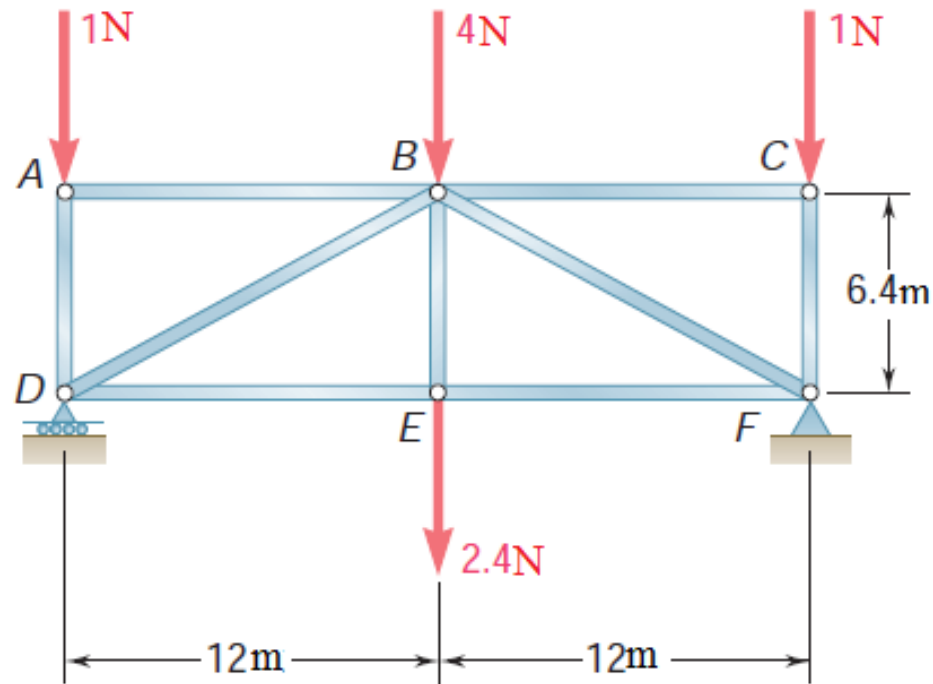
Here, we will **calculate forces acting on the joint, not the forces in the member.**



Analysis of Trusses by Method of Joints

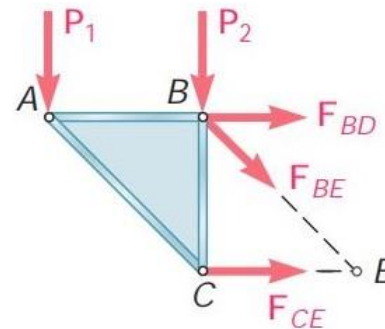
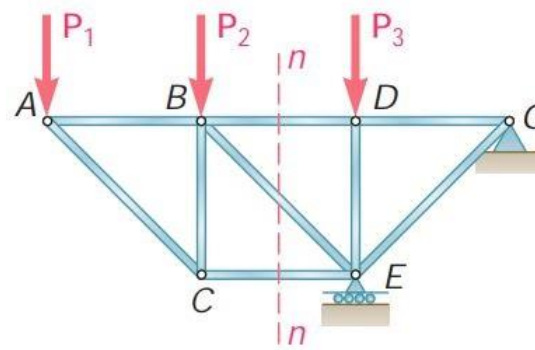
➤ Problem:

Using the method of joints, determine the force in each member of the truss shown. State whether each member is in tension or compression

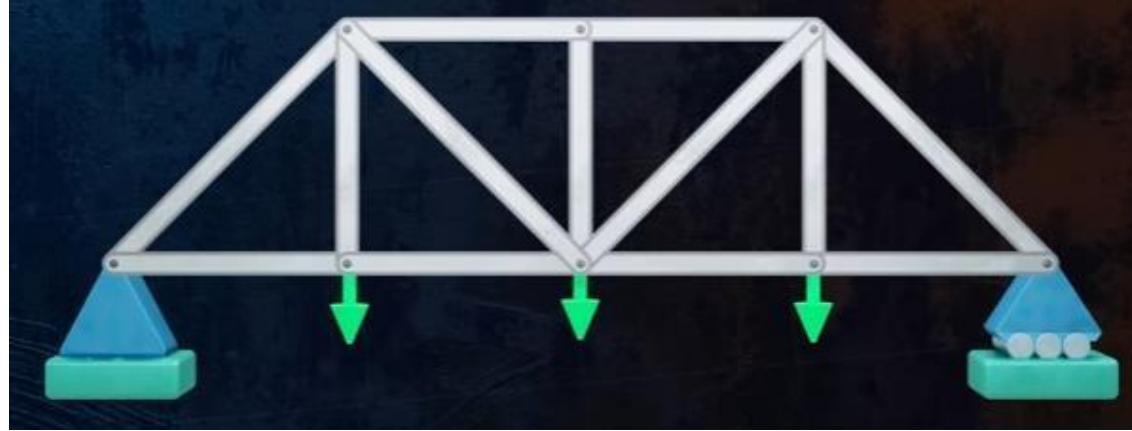


Analysis of Trusses by Method of Sections

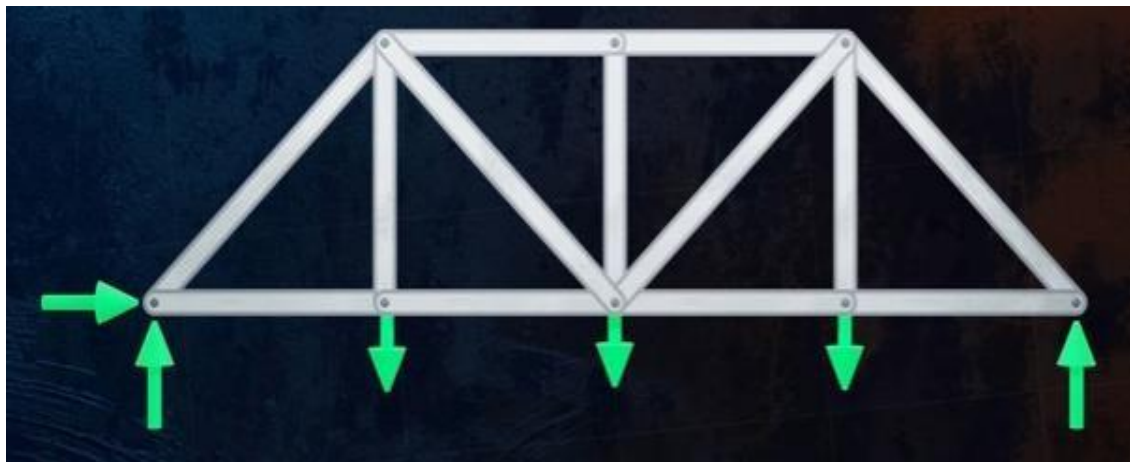
- **Method of Joints** is usually the best method to use when **the forces in all the members of a simple truss** are to be found.
- Method of Sections is **more effective when the force in only one member or the forces in a very few members** of a simple truss are desired.



Analysis of Trusses by Method of Sections



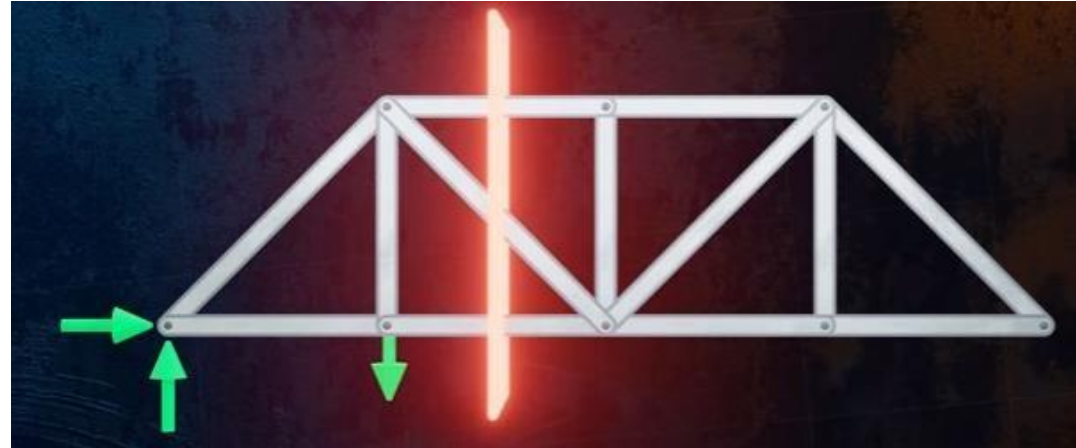
- 1) Draw free body diagram of **the truss** and solve **reaction forces**.



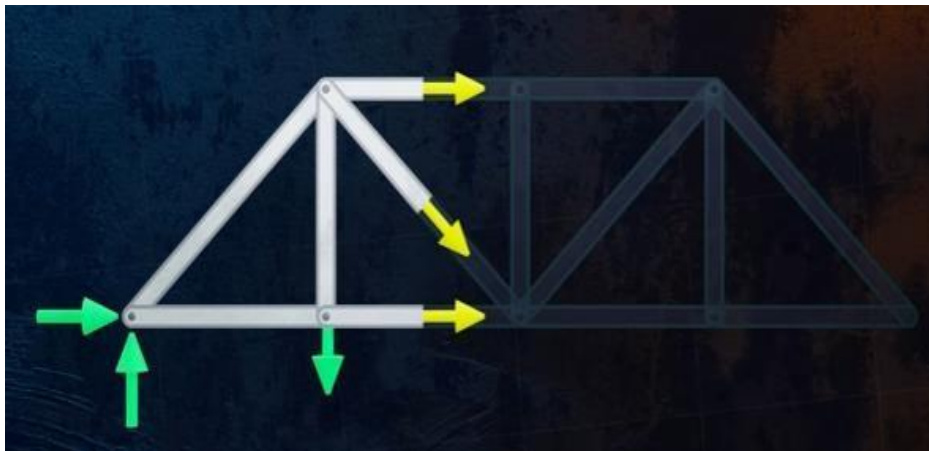
$$\begin{aligned}\sum F_x &= 0 \\ \sum F_y &= 0 \\ \sum M &= 0\end{aligned}$$

Analysis of Trusses by Method of Sections

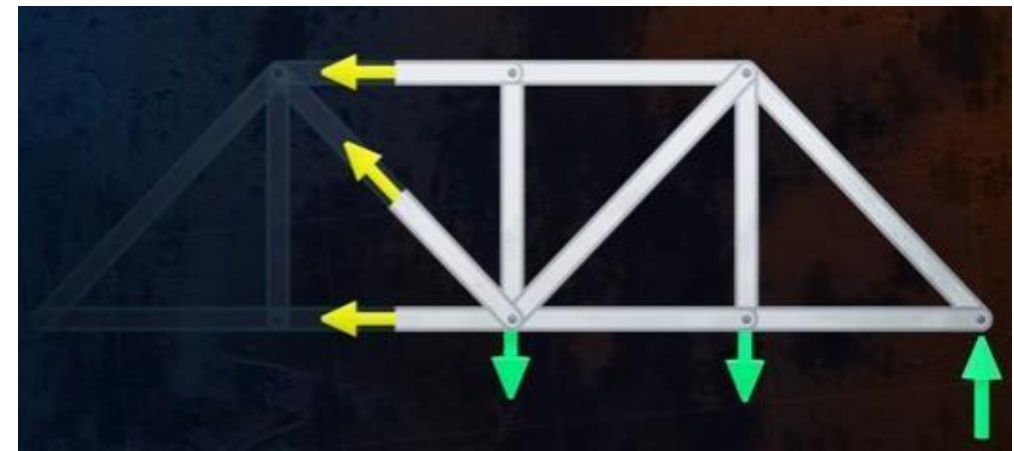
2) Cut the truss through the members of interest and draw the **internal forces**



3) Apply the **equilibrium equations** to solve internal forces



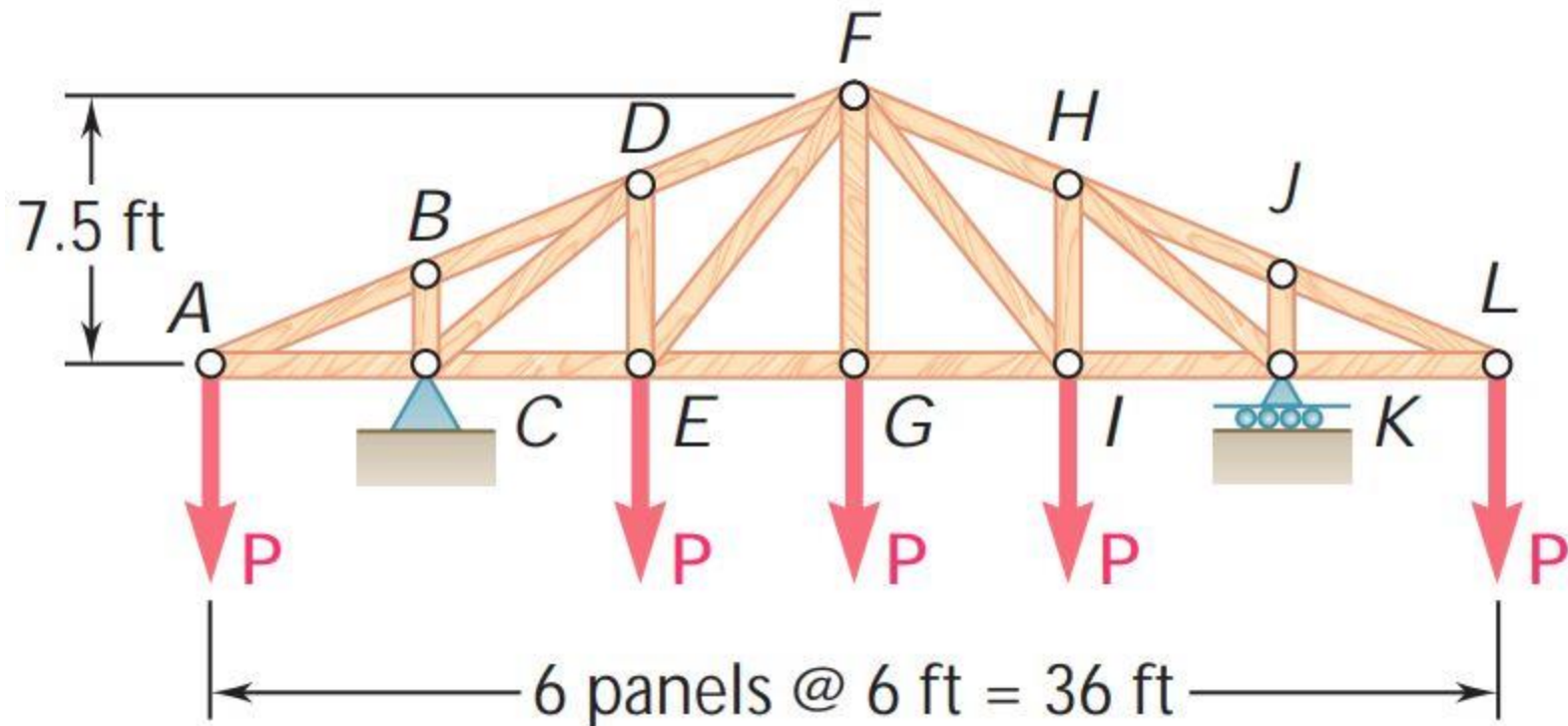
or



Analysis of Trusses by Method of Sections

➤ Problem:

Determine the force in members ***DE*** and ***DF*** of the truss shown when ***P* = 20 kips**



Analysis of Trusses by Method of Sections

➤ Practice Problem:

Determine the force in members ***EH*** and ***GI*** of the truss shown.

(***Hint:*** Use section aa)

